Computació Gràfica i Multimèdia

Work package 4

Master's Degree in Informatics Engineering

2026/2027 year

This document describes the tasks to carry out as part of work package 4.

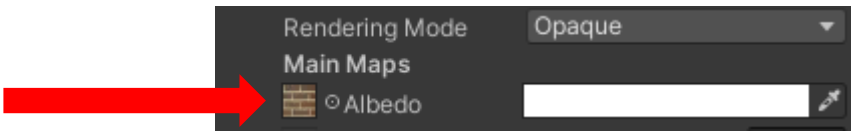
Task 1. Texture mapping

Apply textures to your map. The assets you use for representing the floor and the walls must be textured.

1. Search the Internet for some texture representing a brick wall.



2. Download it and drag it to the assets area of the Unity workspace.
3. Create a new Material and name it "BricksMat". Then drag the texture into the Albedo slot of this material.



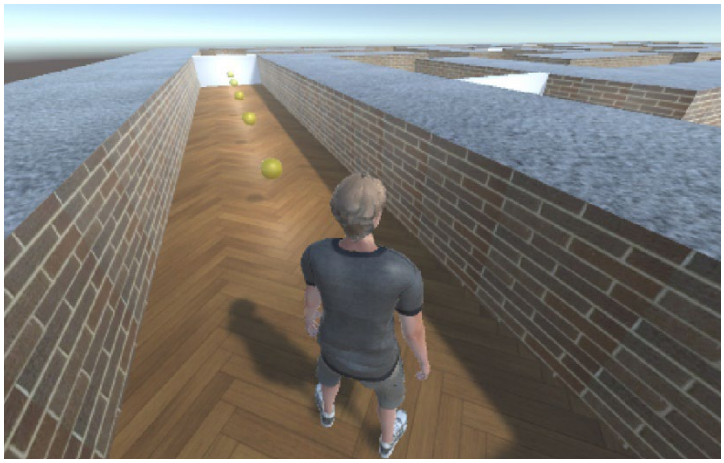
4. Then, drag this material into the Inspector area of your current "Wall" asset (probably a 3D cube).
5. By modifying the "Tiling" parameters you can change the number of times the texture pattern is repeated in the object.



6. In order to use a different texture for some of the cube faces, you will need to create a new asset composed of flat Quad objects and use different materials for each different texture.



7. In work package 4 you are required to create such an asset for its use as a “Wall” object. Its top polygon should be different from the others. If required, modify the asset dimensions so that you can see the top polygon when playing.
8. Choose an appropriate texture for its use in the “Corridor” map object.



Task 2. Game Menu

Note: You can develop this part by following the self-learning document available at the virtual campus.

1. Create a start menu for your project. It should, at least, include three buttons: “Play”, “Credits” and “Quit game”.
2. If you press the “Credits” button you should see information about the game programmer, art director, VFX designer, and so on.
3. When playing the game, the player must be given the opportunity to interrupt the game and go back to the start menu (by pressing the ‘Esc’ key).

Task 3. Pause Menu

1. Add a pause feature to your game. Upon pressing the ‘P’ key, a HUD will appear on the screen. This HUD element includes a button to resume the game.

2. This HUD has to be part of the game scene. You will display it by enabling it. You will make it disappear by disabling it.



You can pause all the elements of the game through the following instruction:

```
Time.timeScale = 0;
```

You can resume the game with:

```
Time.timeScale = 1;
```

Task 4. Shader graph

1. Make use of at least one shader graph in your project.

Evaluation

The mark for this task is computed as:

- 80% : tasks detailed in this document
- 20% : additional contributions of your choice

The additional contributions of your choice should not take more than 20% of the time you devoted to this work package.

Submission

This task is presented as follows:

- Prepare a PDF document containing some snapshots showing that you have properly completed the task and upload it to the activity created in the virtual campus.
- Show your completed work package to the professor who will ask you some questions.